Estimating the Market Value of Attacking Football Players Using Multiple Linear Regression

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Dissertation written under the supervision of Julien Fouquau

Dissertation submitted in partial fulfilment of requirements for the MSc in Finance at CLSBE, at Universidade Católica Portuguesa and for the MSc in Management at ESCP Business School, May 31st, 2022.

Abstract (EN)

The challenging problem of understanding the value of football players inspired countless analyses both in the academic and professional world. This paper has introduced a regression model to estimate attacking football players' market value, building on the findings of prior academic research. The model included performance metrics, personal information metrics, and commercial potential metrics as well. The alternative model introduced was derived from a sample consisting of Premier League attackers, but the regression was later tested on attackers from the Bundesliga as well. Using cross-sectional data, including 105 attacking players from the English Premier League, 27 metrics were exported. All data were retrieved from freely accessible databases (Transfermarkt, FBREF, Instagram). This research paper successfully introduced two new variables not considered by previous academic research: (1) commercial potential and (2) nationality based on the homegrown rule. The proposed alternative regression has an R^2 of 0.65, eliminating multicollinearity and heteroskedasticity, and out-performs the regression models based on prior academic literature, both in-, and out-of-sample. Based on this study, the most important metrics to estimate a football player's market value are (1) expected goals and assists, (2) pressures, (3) player's age, (4) player's nationality, (5) club's prestige, and (6) player's commercial potential.

Key words: Football player valuation | Football player market value | Football club, human capital | Multiple regression model correlation analysis | Multicollinearity | Heteroskedasticity | Performance metrics | Personal information metrics | Commercial potential | Correlation analysis

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Resumo (PT)

O desafio de compreender o valor dos jogadores de futebol tem inspirado inúmeras análises, tanto no mundo académico como profissional. Este documento introduz um modelo de regressão para estimar o valor de mercado de jogadores de futebol atacantes com base nos resultados presentes na literatura académica sobre o tema. O modelo inclui métricas de desempenho, métricas de informação pessoal, e também métricas de potencial comercial. O modelo alternativo introduzido foi derivado de uma amostra constituída por atacantes da Primeira Liga inglesa (*Premier League*), mas a regressão foi posteriormente testada também em atacantes da Primeira Liga alemã (Bundesliga). Utilizando dados transversais, incluindo 105 jogadores atacantes da Primeira Liga inglesa, foram exportadas 27 métricas. Todos os dados foram extraídos de bases de dados de livre acesso (Transfermarkt, FBREF, Instagram). A investigação introduz com sucesso duas novas variáveis não consideradas pela investigação académica anterior: (1) potencial comercial e (2) nacionalidade com base na regra de jogadores formados localmente. A regressão alternativa proposta tem um R^2 de 0,65, eliminando problemas de multicolinearidade e a heterocedasticidade. Como tal, supera os modelos de regressão baseados na literatura académica anterior, tanto dentro, como fora da amostra. Com base neste estudo, as métricas mais importantes para estimar o valor de mercado de um jogador de futebol são (1) golos e assistências esperadas, (2) pressões, (3) idade do jogador, (4) nacionalidade do jogador, (5) prestígio do clube, e (6) potencial comercial do jogador.

Palavras-chave: Avaliação de jogadores de futebol | Valor de mercado de jogadores de futebol | Clube de futebol, capital humano | Análise de correlação do modelo de regressão múltipla | Multicolinearidade | Heteroskedasticidade | Métricas de desempenho | Métricas de informação pessoal | Potencial comercial | Análise de correlação

Título: Estimar o valor de mercado de jogadores de futebol que atacam utilizando a Regressão Linear Múltipla

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1 Introduction

The rapidly growing economization of football makes football players' valuation a hot topic nowadays. As football has grown into a multi-billion business today, many companies have developed sophisticated solutions to help scouts and managers value their selected players more efficiently. The topic also inspired many academic researchers. Through empirical findings, numerous studies have attempted to identify the key metrics influencing football players' market value. However, as the influence of data analytics has increased rapidly over the past decades, the value of data increased exponentially. Professional data providers are costly, while the limited data available on free databases is hard to collect, as these databases are primarily made for football enthusiasts. The constraint of only accessing free databases limits the academic papers' efficiency.

This paper attempted to present a regression model to estimate forward players' market value with minimizing the inefficiency sourced from the data gathering constraint. Using a data scraping code written by the author in Python to scrape the free football statistics provider FBREF's website made it possible to examine twenty different performance measures and choose the most critical variables that affect players' market value. The sample was set up using data of all attacking players from the English Premier League. The database was constructed from three different sources (1) Transfermarkt, (2) FBREF, and (3) Instagram. The most recent data up to April 1st, 2022, was considered.

Using STATA as the statistical software to perform all calculations, the paper analyses two different regression models inspired by previous academic research and proposes an alternative regression model with two new variables to improve estimation accuracy further. The newly introduced variables in the alternative regression capture a significant amount of the players' market value, outperforming the two regressions inspired by the prior academic literature.

In-sample and out-of-sample tests are run to test the estimating power of all regression models discussed in the paper, using all 105 English Premier League attackers as the in-sample players and randomly selected 20 German Bundesliga attackers as the out-of-sample players.

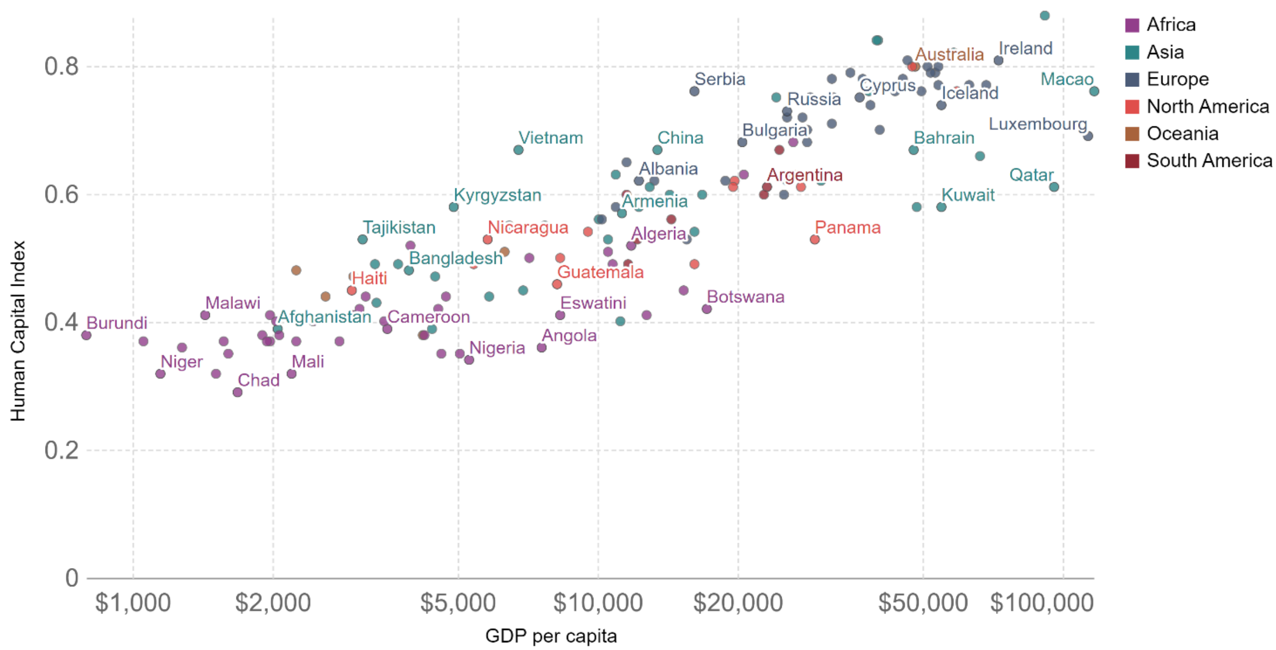
2 Literature Review

2.1 Human Capital

The knowledge economy is built on assembled, talented, and trained workforces. The economic value of the assets directly related to the workforce, such as education, training, and intelligence, is called Human Capital. (Merriman, 2017) The importance of human capital is unquestionable in modern business. As Gamerschlag (2013, p. 327) states: “Against the background of human capital theories and the resource-based view of the firm, human capital must be regarded as a central factor behind organizations’ competitiveness.”

The economic theory always suggested a positive relationship between human capital and economic growth, and this contribution is now well documented through countless research studies. For example, Pelinescu, 2014 found a statistically significant relationship between GDP per capita and the innovative capacity of human capital (evidenced by the number of patents) and the qualification of employees (secondary education). Furthermore, a 2004 report based on EU statistics measuring GNP, unemployment, and various educational and vocational training investments suggested that investing in human capital (through education and training) can be associated with economic growth across most member states. (Rob A. Wilson, 2004)

Figure 1: Human Capital Index vs. GDP per capita in 2016



Source: Our World in Data, 2022, https://ourworldindata.org/grapher/human-capital-index-vs-gdp

As shown in Figure 1, there is a positive relationship between the Human Capital Index (HCI)[1] and the GDP per capita, just as Pelinescu, 2014 suggested. Generally, a higher HCI in a country suggests a higher GDP per capita.

Although the importance of human capital is well established, only a few organizations evaluate their workforce in economic terms since human capital is not disclosed on the assets side of the classical balance sheet. Organizations do not actively value human capital because they do not own human companies; employees do. (Investopedia, 2022) All related cash outflows (salaries, etc.) are recorded as expenses in the income statement. (Accounting Tools, 2022) Hence, companies have no obligation to continuously disclose their human capital's value, contrary to

[1] Human Capital Index (HCI): "combines indicators of health and education into a measure of the human capital that a child born today can expect to obtain by his/her 18th birthday, given the risks of poor education and health that prevail in the country where she lives. The HCI is measured in units of productivity relative to a benchmark of complete education and full health, and ranges from 0 to 1." (Our World in Data, 2022)

their other assets. Nevertheless, data and analytical capabilities available to measure the value of human capital are more available than ever before, and the topic inspires researchers.

2.1.1 Valuation of Human Capital in Academic Research

The book by Merriman, 2017 titled "Valuation of Human Capital," describes the shifting perspective of human capital over time, but it also provides an overview of all the approaches professionals use to value human capital. The same valuation approaches used to value most asset types (such as businesses or real estate) could also be used to value a firm's aggregate human capital. These approaches are:

- cost approach valuation,
- income approach valuation,
- market approach valuation.

The cost approach is based on the economic principle of substitution, which tries to answer the question: how much would it cost to substitute the asset? Likert, 1971 was the first to propose a valuation method based on the cost approach for human capital. The replacement cost method is based on how much it would cost to replace existing human resources. The model considers costs associated with finding, hiring, training, and developing human resources.

According to Merriman 2017, the income approach evaluates the human capital required based on the expected net income stream directly attributable to the workforce. One of the first studies on the topic was published in The Journal of Political Economy in 1961. The paper has attempted to indicate the importance of developing an estimate of human capital and discuss some difficulties. Weisbrod, 1961 used the present value of all future earnings to value an employee's value.

The market approach calculates the value by looking at the selling prices of the asset or a similar asset. In their paper titled "The Role of Human Capital in Firm Valuation: An Application on BIST," Cam & Özer, 2016 try to find human capital's explanatory power and value relevance in the Ohlson Model's residual. They have found that human capital is indeed value-relevant in making business decisions and that the value of human capital can explain a significant part of the residual in the sample (Turkish, listed companies).

2.1.2 Valuation of Human Capital in Practice

According to the 2016 Global Intangible Financial Tracker (GIFT) report, intangible assets gave more than 50% of companies' market value. (Haigh, 2017) Since human capital's contribution to a company's market value can be so massive and because of the increased emphasis on employee well-being, the measurement of the value of human capital was a central topic at the World Economy Forum, 2020. Two main goals were worded at the conference:

- the current human capital valuation framework is not complete and insufficient,
- stakeholders should create a framework for monitoring and assessing the return on investment made in their workforce.

In recent years, efforts to include human capital in financial reports have intensified due to a clear and rising market interest in understanding how corporations manage and assess human capital to meet all stakeholder interests.

In collaboration with Willis Tower Watson, the World Economic Forum has published a new paper called "Human Capital as an Asset: An Accounting Framework for Reset the Value of Talent in the New World of Work," which aims to reshape human capital accounting in organizations globally. The framework allows the organization to calculate a return on investment for its employees and aims to be a guide to help organizations efficiently adopt the new framework. The paper includes seven guiding principles to shift how human capital is valued today.

Table 1: The steps to shift how human capital is valued in organizations

Steps	From		To
1)	**Profit** Value for a narrow group of stakeholders	→	**Purpose** Shared value for a broad group of stakeholders
2)	**Corporate Policy** Complying with code of conduct	→	**Social Responsibility** Living corporate values in the community
3)	**Stand-alone** The organization as a stand-alone entity	→	**Ecosystem** Organization as an integral part of the communities
4)	**Employees and jobs** Process-centric: hire people for fixed roles	→	**People, work and skills** Human-centric: Empowering talent with meaningful work
5)	**Workforce as an expense** Treating talent as business expense	→	**Workforce as an asset** Valuing talent as an asset
6)	**Backward-looking financial metrics** Focusing on past financial performance	→	**Forward-looking financial metrics** Focusing on future pontetial of value creation
7)	**Quarterly** Short-term view	→	**Generational** Time agnostic

Source: Author; Adapted from: World Economic Forum, 2020, https://www3.weforum.org/docs/WEF_NES_HR4.0_Accounting_2020.pdf

As we can see, human capital valuation is becoming a crucial topic worldwide. Both scholars and business professionals are looking for a standardized way to value human capital precisely. The relevance of the topic nowadays is unquestionable.

2.2 Football Players

Human capital is essential to a professional sports team's value, like most organizations. Sports teams' value highly depends on their performance as good performance generates more revenue in many ways and good performance is only achievable with outstanding human capital, or in this case, with outstanding football players.

2.2.1 Football Players as Human Capital

Many would agree that the people make an organization in the digital age, especially in football, where players' performance is closely tied with the football club's value creation. Hence, in football, the valuation of human capital is fundamental. Based on Perechuda & Cater, 2021, a clubs' players' valuation is essential for a football club's value creation. 62% of football clubs' success is associated with intellectual capital, including player contracts.

Furthermore, costs associated with players are a significant component of clubs' overall expenses. Based on the study of Perechuda & Cater, 2021, on average, clubs payout 86% of their revenues as salary, a significant portion of these salaries being player salaries. The high salaries are to attract more valuable players. More valuable players escalate higher broadcasting deals, sponsorship deals, commercial merchandise sales, and income from transfer fees. Overall, a football club's revenue and value increase with more valuable players. But how do clubs value these players? What are the most critical factors in a football player's market value?

Finally, it is important to note that accounting-wise, football players are treated differently compared to other employees since players are revenue-generating assets and can be directly associated with revenues. When a team signs a player, it earns the right to use the player to generate revenue, so the contract is considered an intangible asset. (RSM, 2017)

2.2.2 Valuation of Football Players in Academic Research

Plenty of academic literature exists concerning the valuation of professional football players, which is based on empirical findings. Data in the world of professional sport is extremely valuable; thus, without subscribing to a professional data provider, the data available for such research is limited and very hard to collect.

Stanojevic & Gyarmati, 2016 produced a method to assess football players' market value mainly based on performance measures. They considered Transfermarkt estimates of the market value equal to the actual market value perturbed with a noise factor. Stanojevic & Gyarmati, 2016 only used performance indicators to estimate market values because the paper's end goal was to find a relationship between squads' market value and game outcomes. Not surprisingly, a performance-based model estimates outcomes more precisely, proving that performance measures significantly affect players' market value. Using multiple databases, the researchers measured more than 30 performance-related variables' effects on a player's market value.

On the contrary, Metelski, 2021 used actual transfer values in his study. The study aimed to show a relationship between transfer values and players' age. Even though Metelski, 2021 found a significant negative relationship between the transfer value of players and their age, the model could not be used to estimate market value. The lack of precise data limits researchers' ability to produce more sophisticated models using actual transfer values. Also, it is essential to note that even the so-called “real-time transfer values” are mainly estimations. Football clubs have no obligation to publish transfer values; thus, these values are usually just media estimates.

Players’ market value factors identified by previous studies can be divided into two groups: team factors (league, FIFA ranking, Champions League qualification, etc.) and individual factors. Individual factors can be further broken down into personal information (height, weight, age, position) and performance measures (goals, assists, etc.).

For example, Majewski, 2016, found that the variables that have the most effect on a forward player’s value from the top 150 most valuable footballers in the world are:

- individual factor: Canadian classification point (number of goals + number of assists)
- team factors: the value of the players club adjusted by the FIFA ranking points and the factor representing “goodwill” or “brand” of the specific player.

The study of Wicker, 2013 assumed that individual effort could be a factor in players’ market value. However, interestingly enough, the data showed that effort (measured by intensive runs and total distance runs per game) had no significant impact on football players’ value.

Intuitively, one would think that since players’ performance greatly influences a player’s market value, different position players would be valued differently. Nsolo, Lambrix, and Carlsson, 2018 confirmed that this hypothesis is true. Similar methods have different results for different positions. The paper has distinguished four positions:

- Goalkeepers
- Defenders
- Midfielders
- Attackers

Table 2: List of the last twenty Ballon d'Or winners

Year	Ballon d'Or Winner	Position
2021	Lionel Messi	Attacker
2020	*Canceled due to the COVID pandemic*	-
2019	Lionel Messi	Attacker
2018	Luka Modrić	Midfielder
2017	Cristiano Ronaldo	Attacker
2016	Cristiano Ronaldo	Attacker
2015	Lionel Messi	Attacker
2014	Cristiano Ronaldo	Attacker
2013	Cristiano Ronaldo	Attacker
2012	Lionel Messi	Attacker
2011	Lionel Messi	Attacker
2010	Lionel Messi	Attacker
2009	Lionel Messi	Attacker
2008	Cristiano Ronaldo	Attacker
2007	Kaká	Midfielder
2006	Fabio Cannavaro	Defender
2005	Ronaldinho	Attacker
2004	Andriy Shevchenko	Attacker
2003	Pavel Nedvěd	Midfielder
2002	Ronaldo	Attacker
2001	Michael Owen	Attacker
2000	Luís Figo	Attacker

Source: Author; Adapted from: Wikipedia, 2021, https://en.wikipedia.org/wiki/Ballon_d%27Or

Attacking players play an essential role in research. This could be justified by the fact that although attacking players only represent 30% of all players, they are the most valued players, as we can see from the Ballon d'Or awards[2]: attacking players won 16 out of the last 20 awards (3 midfielders and 1 defender has won the award, while no goalkeeper has ever got one). (Wikipedia, 2022)

Kologlu et al., 2018 used a multiple linear regression approach to estimate attacking players' market value. Although the researchers could not eliminate multicollinearity, they came up with a partially successful model. Their most interesting findings were that players' age has a significant negative relationship with market value, players between the height of 180 cm and 184 cm have

[2] The Ballon d'Or is an annual football award presented by French news magazine France Football since 1956, although between 2010 and 2015, the award was temporarily merged with the FIFA World Player of the Year. (Wikipedia, 2022)

the highest market value, and that yellow and red cards have no significant effect on players' market value.

The paper of Li, 2020 is similar to Kologlu et al., focused solely on the English Premier League clubs and built a multiple regression model to predict football players' value. Li, 2020 used 14 different independent variables, 4 of them labeled as personal information and 10 as performance metrics. Out of the 14 variables, Li, 2020 found a significant positive effect for goals scored and average time spent on the field per game and an adverse effect for players' age and club rank. Surprisingly, other variables such as the average number of assists or number of yellow cards received had no significant impact on the market value.

Interestingly enough, no such academic paper was found that attempted to connect a player's popularity (and thus commercial potential) with the player's market value. Multiple papers considered player goodwill or player brand a factor, such as Majewski, 2016, but the goodwill was defined as the residual value between the predicted market value (from team and player characteristics and performance) and the "real market value." However, no paper tried to use an independent variable to proxy a player's popularity in a predicting model.

2.2.3 Valuation of Football Players in Practice

As football has grown into a multi-billion-euro business, multiple companies have developed sophisticated solutions to help scouts and managers value their selected players more efficiently.

One of the most well-known businesses in this field is Ace Advisory's Football Benchmark. Football Benchmark (FB) has a valuation tool based on a linear regression model (generalized linear model) to identify the most relevant variables that influence a player's value. (Ace Advisory, 2022). The model is based on past football player transactions, and the data is provided by leading sports data company Hudl/Wyscout. FB identified several drivers as key to estimating a player's value. Most important drivers:

- player's position,
- player's age and nationality,
- player's contractual situation,
- individual performance measures,

- media and commercial potential,
- team performance and characteristics,
- selling team's dependence on the player,
- timing of the transfer

Interesting to note that FB considers players' marketing/brand value. Also, besides a study conducted by Dobson, Howe, and Gerrard, 2020 on English nonleague football players, no other academic research has considered players' contractual status. Dobson, Howe, and Gerrard found that the expiration date of a player's contract is likely to have a massive impact on a player's value, especially if his contract runs out in the next 12 months.

Similar to FB, CIES Football Observatory has a statistical model to estimate the market value of football players. The model is built on 1 790 paid transfers between July 2012 and January 2020. The variables used to predict transfer values are very similar to those of FB, and the correlation between sums estimated and paid is almost 85%, based on their March 2020 report. (CIES, 2020)

Table 3: Comparing Football Benchmark and CIES market value estimations

Player (Age)	**Club**	**Nationality**	**Market value (€m)**	
			Football Benchmark	**CIES**
Haaland (21)	Dortmund	Norway	138	90-120
Foden (21)	Manchester City	England	125	120-150
Alexander-Arnold (23)	Liverpool	England	111	90-120
Bruno Fernandes (27)	Manchester Utd	Portugal	110	120-150
F. de Jong (24)	Barcelona	Netherlands	109	90-120
Salah (29)	Liverpool	Egypt	108	50-70
Kane (28)	Tottenham	England	107	50-70
Sancho (21)	Manchester Utd	England	106	90-120
Lukaku (28)	Chelsea	Belgium	105	50-70
R. Dias (24)	Manchester City	Portugal	104	120-150

Source: Author; Adapted from: Football Benchmark, 2022, https://www.footballbenchmark.com/home; CIES, 2022, https://football-observatory.com/-values-126630

Since FB and CIES have a very similar methodology, Table 3 compares the top 10 most valuable (according to FB) players' values. As shown in Table 3, CIES gives a range of values, while FB provides a specific value. Although both data providers use a similar methodology, there is a significant difference in results. One possible explanation could be that the two data providers give players’ ages a different weight when calculating their market value. There is a big difference between the two valuations for all three players that are 28 or above. CIES likely has a more considerable discount for older players than FB. However, to test this hypothesis would need more than just limited (free) access to both of these datasets.

With the developments in the field, artificial intelligence companies also build models that enable more innovative recruitment and player development decisions. The British company Ai Abacus, for example, uses predictive models based on artificial intelligence not only to predict players’ market value but also to predict how well a specific player will fit a specific club’s playing style and dynamics of the team. (AI Abacus, 2022)

Based on an extensive academic and professional literature review, two standard methods exist for football players’ market value regression models. The first method is to solely rely on actual transfers and assume the transfer fees to be the market value of a given player on the given date when the transfer happened. Professional companies typically use this method (such as Football Benchmark). While using actual transfer fees as the independent variable has its advantages, it also has limitations. The main limitation is that the actual transfer values are still just media estimates since the accurate figures are not published by the clubs involved in most cases.

Popular in academic research, the second method uses a proxy as the players’ market value. Without a professional data provider and relying only on public data, it is challenging to extract football players for a given date (transfer date). Also, by using a proxy (such as the values published by Transfermarkt), researchers can increase the sample size, which, all else equal, increases a model’s reliability.

2 Research design

2.1 Database Review

2.1.1 Database construction

As mentioned before, it is very challenging to find reliable databases from which it is easy to extract the data. For this research, data were collected from three different websites:

- Transfermarkt

- FBREF
- Instagram

Like many other researchers, this paper also uses Transfermarkt as the primary data source. The market value of the players and other personal information, such as age and nationality, were all extracted from Transfermarkt. (Transfermarkt, 2020) While the website does not have an option to export pre-sorted data, player data can be sorted by teams, and from there, it is easy to copy the data to an excel spreadsheet.

Since only one observation was needed per player from the social media platform (number of followers), the data extraction was conducted case-by-case and by hand, similarly to the Transfermarkt data. The main challenge was to find all 120 players' official Instagram accounts. However, Instagram's account verification system made identifying the players' accounts easier. Interestingly enough, only one player out of the 120 players did not have an Instagram account: Ashley Barnes from Burnley. Moreover, only one other player had a non-verified Instagram account, Przemyslaw Placheta, from Norwich City.

The most challenging part of the data construction was gathering all the performance measures. Out of the accessible databases, online FBREF proved to be the most precious. FBREF does not only have all sets of performance measures, from attacking measures (such as goals scored) to defending measures (tackles made), it also provides predicted measures (expected goals) and has all these values in a per 90-minute form (for example goals scored per 90 min), which makes the observations standardized and comparable.

However, it is difficult to extract data from FBREF. Only one player can be examined at a time. Due to this crucial constraint, a data scraping code was developed in Python[3] to automate the data extraction from FBREF. This method made the data construction more efficient and made the database easily updateable over time. The code used to scrape FBREF is attached in the appendix (Appendix 1). Hopkins, 2021 provided a practical case study that helped create the scraping code.

[3] Python is a high-level, general-purpose programming language (Wikipedia, 2022)

All data were collected and organized in an excel spreadsheet. After having the final version, the database was imported into STATA, a general-purpose statistical software. From there on, all data manipulation was conducted within Stata software using its econometric panel models.

Overall, the dataset consists of the 120 players categorized as attacking players on Transfermarkt. After eliminating players that do not have sufficient game time in any of the top 5 leagues (Premier League in England, La Liga in Spain, the Bundesliga in Germany, Serie A in Italy, and Ligue 1 in France), 105 players were left in the dataset. 28 different variables from 3 different sources were extracted to form the final database for all players.

2.1.2 Personal information from Transfermarkt

Personal information variables from Transfermarkt:

Team: Attacking player's current team. Players are included from all 20 Premier League clubs (based on season 2021/2022).

Market value: On Transfermarkt, the Market Values are calculated by a consensus from approved and confident experts on the website. Although this method seems subjective, previous academic research (Wicker, 2013) suggests it works well as a real-time market value proxy for football players.

Position: central forwards (CF), right-wing (RW), and left-wing (LW).

Age: Current age of the player.

Nationality: Nationality of the player. Suppose the player has double nationality; only the first one is considered.

Contract length: How many years are left from the player's contract (in years).

Foot: Players dominant foot is left, right, or both.

2.1.3 Commercial potential measures from Instagram

Commercial potential measures from Instagram:

Followers: Number of Instagram followers, in thousands

2.1.4 Performance measures from FBREF

Performance measure variables from FBREF:

Non-penalty goals: Non-penalty goals scored per 90 minutes in the last year.

npxG: Expected non-penalty goals are the probability that a non-penalty shot will result in a goal based on the characteristics of that shot and the events leading up to it. It is also used to check how many goals are expected from a player in each game. The expected goals are calculated based on FBREF's methodology (FBREF, 2022).

Shots total: Total shot attempts per 90 minutes in the last year.

Assists: Total assists per 90 minutes in the last year.

xA: Expected total assists per 90 minutes in the last year. The expected assists are calculated based on FBREF's methodology, similar to the expected goals.

npxG + xA: Sum of expected non-penalty goals and expected assists.

Shot creating actions: Includes live-ball passes, dead-ball passes, successful dribbles, shots which lead to another shot, and being fouled, which leads to a shot.

Passes attempted: Number of passes attempted per 90 minutes in the last year.

Pass completion: Average pass completion in % per 90 minutes in the last year.

Progressive passes: Number of progressive passes per 90 minutes in the last year.

Progressive carries: Number of progressive carries per 90 minutes in the last year.

Dribbles completed: Number of dribbles completed per 90 minutes in the last year.

Touches: Number of touches in the opposition's penalty box per 90 minutes in the last year.

Progressive passes received: The number of progressive passes received in the opposition's penalty box per 90 minutes in the last year.

Pressures: Total times player has put pressure on the opposite team per 90 minutes in the last year.

Tackles: Total tackles per 90 minutes in the last year.

Interceptions: Total interceptions per 90 minutes in the last year.

Blocks: Total blocks per 90 minutes in the last year.

Blocks: Total clearances made per 90 minutes in the last year.

Aerials won: Total aerial duels won per 90 minutes in the last year.

2.2 Descriptive statistics

2.2.1 Personal information metrics

Out of the 8 personal information extracted from Transfermarkt, only 4 are numerical variables, and the other 4 are categorical variables.

Table 4: Descriptive statistics of market value, age, height, and contract length

Variable	Obs	Mean	Std. Dev.	Min	Max
Market Value	106	24.1	22.7	1	100
Age	106	26.2	3.8	19	36
Height (cm)	106	180.0	6.2	163	197
Contract Length (Y)	106	2.1	1.4	0.1	5.1

Source: Author

Analyzing Table 4, we can conclude that an average attacking player's market value in the English Premier League is €24.1 million. The cheapest player is valued at €1 million (Aaron Lenon, Burnley), while the most expensive players are valued at €100 million (Harry Kane, Tottenham & Mohamed Salah, Liverpool).

Furthermore, the average player in the database is 26 years old and 180 cm tall. Although the average contract length is a little above two years, there are several players with contracts running out within a year.

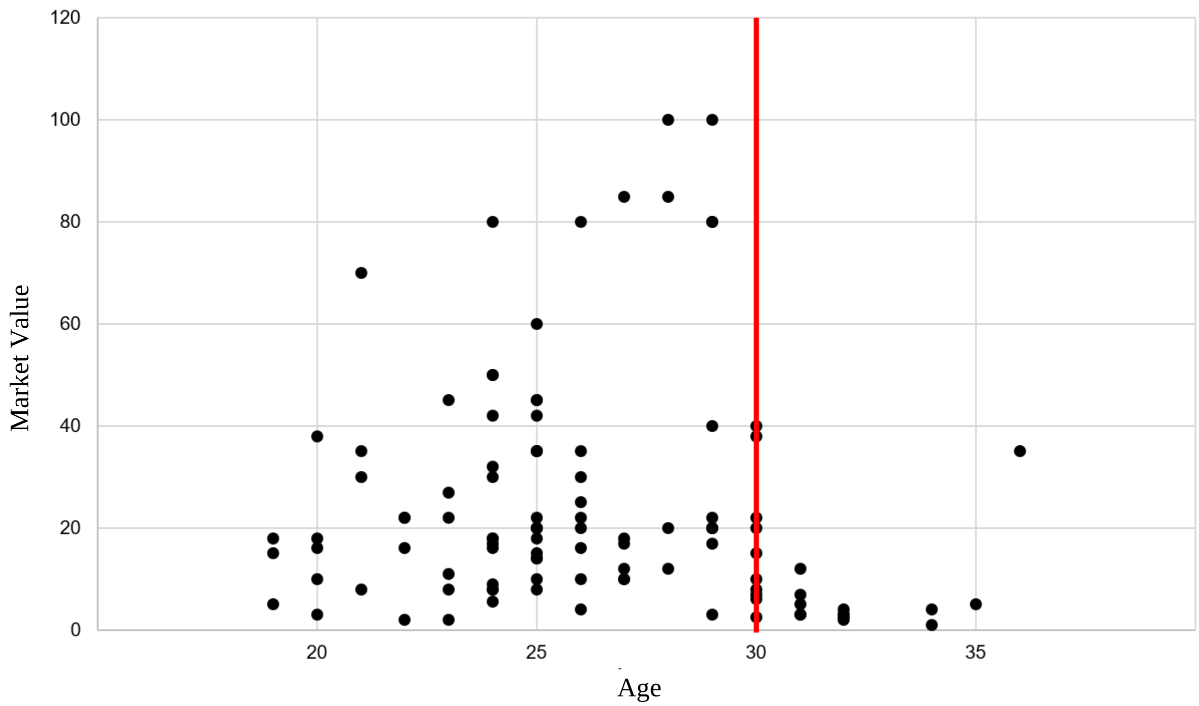
Table 5: Correlation between market value and personal information metrics

	Market Value
Age	-0.0834
Height (cm)	0.000404
Contract Length	0.208*

*Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; Source: Author*

Although even with a 5% confidence level, only contract length seems to correlate with market value significantly, it is worth attempting to convert all three variables into categorical variables. For example, age as a numerical variable might not influence players' market value; however, theory suggests that players above a certain age lose market value.

Figure 2: Scatter plot graph between market value and age



Source: Author

As we can see from the Scatterplot graph in Figure 2, although there is no linear relationship between market value and age, all players' market value for players above 30 is considerably lower.

Regarding the categorical variables, nationality and the team variables were converted to dummy variables as discussed in the Empirical Methodology part.

Table 6: Descriptive statistics for nationality, team, position, and foot variables

Nationality			Big 6		
	Frequency	**Share (%)**		**Frequency**	**Share (%)**
English or Welsh	32	30%	Big 6 club's player	27	25%
Other	74	70%	Other club's player	79	75%
Position			**Foot**		
	Frequency	**Share (%)**		**Frequency**	**Share (%)**
Central Forward	31	29%	Right	82	77%
Left Wing	49	46%	Left	22	21%
Right Wing	26	25%	Both	2	2%

Source: Author

Previous academic research has established that teams' prestige and position influence players' market value. However, based on countless test regressions, further breaking down positions from the first level (attackers) into a subgroup has no significant effect on market value. Also, the market value is not sensitive to the players' dominant foot among attackers at the English Premier League level.

Thus, the position and foot variables have been dropped, while the nationality and big six variables will be further used as dummy variables in the regression models.

2.2.2 Commercial potential metrics

Table 7: Descriptive statistics of follower numbers

thousand of followers

		Percentiles	
Obs	106.0		
Mean	6 466		
Std. Dev.	41 682	**25%**	119
Skewness	9.8	**50%**	348.5
Kurtosis	99.6	**75%**	1 600

Source: Author

As shown in Table 7, the median (349 thousand) is well below the average (6.5 million). The number of followers has a positive skewness and kurtosis, which suggests that while most of the player's follower base is well below the average, a few extremely high values drive the mean up. Hence, categorizing players based on their follower numbers is more reasonable to capture commercial potential as a market value driver.

2.2.3 Performance metrics

Overall, twenty performance metrics were extracted from FBREF. All metrics are per 90 minutes and based on the last 365 days, making all variables comparable through the dataset. However, it is not realistic to use all twenty performance metrics since the high multicollinearity between them.

First, it is essential to check whether the performance metrics have any significant relationship with the market value to select the performance metrics that could capture some part of players' market value. Then the question of multicollinearity[4] arises, as undoubtedly, there is a relationship between the performance metrics. Although there is no general rule for the cut-off value, a cut-off value of 0.3 is used in this scenario because of the high expected correlation between the variables. For example, expected assists (xA) strongly correlate with assists. The correlation table, including all variables, was analyzed to decide which to keep and drop.

[4] In statistics, multicollinearity is a phenomenon in which one predictor variable in a multiple regression model can be linearly predicted from the others with a substantial degree of accuracy. The existence of multicollineartiy In this situation, the coefficient estimates of the multiple regression may change erratically in response to small changes in the model or the data. (Wikipedia, 2022)

Table 8: Performance metrics' correlation with Market Value

	Performance Metric	Market Value
1	Non-Penalty Goals (NP Goals)	0.364***
2	Expected NP Goals (xnpG)	0.385***
3	Total Shots	0.357***
4	Assists	0.235*
5	Expected Assists (xA)	0.577***
6	Expected NP Goals + Expecte Assists (xnpGA)	0.551***
7	Shot Creating Actions	0.546***
8	Passes Attempted	0.367***
9	Pass Completion	0.345***
10	Progressive Passes	0.352***
11	Progressive Carries	0.419***
12	Dribbles Completed	0.243*
13	Touches in Attacking Penalty Box	0.560***
14	Progressive Passes Received	0.554***
15	Pressures	-0.286**
16	Tackles	-0.265**
17	Interceptions	-0.202*
18	Blocks	-0.189
19	Clearances	-0.248*
20	Aerials Won	-0.175

*Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; Source: Author*

Starting with the defensive performance measures (15-20), Table 8 shows us that all six measures have an inverse relationship with the market value. These inverse relationships were expected, as the more an attacking player is involved in defensive duties, the higher the chance that the player gets injured, which decreases his market value. Moreover, only the correlation coefficient of tackles and pressures is statistically significant; therefore, we drop the other four defensive measures. As one would think, pressures and tackles have a high and statistically significant correlation coefficient of 0.548. To minimize the risk of multicollinearity in the model, we only keep pressures as an independent variable since the correlation between market value and pressures is higher than the correlation between tackles and market value.

Regarding the attacking performance measures' (1-14) correlation coefficients and market value, we can conclude that all fourteen have a positive relationship with the market value, and twelve of them are statistically significant with a confidence level of 1%. Thus, we only drop the variable

Assists and Dribbles completed. Since this study only examines attacking players, dropping completed makes sense as this measure is more critical for midfielders than attackers. Assists are more interesting, as it is one of the essential measures in football. However, we can see a strong and statistically significant relationship between expected assists (xA) and the market value, which could be explained by the fact that assists are not an individual measure per se. Assists rely on others' (teammates') performance, while xA indicates a player's ability to set up scoring chances without relying on the actual result of the teammates' shot. Therefore, one could argue that by eliminating others from the equations, expected assists capture a player's ability to create chances for their team more efficiently than the measure assist.

Table 9: Correlation matrix for attacking measures

	NP Goals	npxG	Shots Total	xA	npxGxA	Shot Creating Actions	Passes Attempted	Pass Completion	Progressive Passes	Progressive Carries	Touches Att Pen	Progressive Passes Rec
NP Goals	1											
npxG	0.779***	1										
Shots Total	0.547***	0.716***	1									
xA	0.275*	0.254*	0.266**	1								
npxGxA	0.749***	0.921***	0.693***	0.608***	1							
Shot Creating Actions	0.0549	0.0176	0.247*	0.719***	0.305**	1						
Passes Attempted	0.047	-0.124	0.0328	0.513***	0.103	0.610***	1					
Pass Completion	0.353***	0.268**	0.183	0.387***	0.375***	0.426***	0.506***	1				
Progressive Passes	-0.0429	-0.15	0.103	0.544***	0.0918	0.714***	0.777***	0.351***	1			
Progressive Carries	-0.013	-0.164	0.058	0.474***	0.0561	0.717***	0.770***	0.487***	0.688***	1		
Touches Att Pen	0.572***	0.687***	0.615***	0.339***	0.698***	0.321***	0.106	0.425***	0.0701	0.209*	1	
Progressive Passes Rec	0.577***	0.618***	0.576***	0.422***	0.676***	0.342***	0.232*	0.451***	0.203*	0.215*	0.781***	1

*Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; Source: Author*

The correlation table in Table 9 shows that the expected assists metric has a significant relationship with all other metrics with a 1% confidence level, except for the expected non-penalty goals (npxG). Since npxG represents how much goal is expected from a player per game based on countless other performance measures. Similarly, xA is calculated from the location of the passer, the type of pass, and the type of attack, making the npxG and the xA measures include almost all other attacking performance measures. Therefore, out of the 16 attacking measures, only the expected non-penalty goals (npxG) and expected assists (xA) are kept as independent variables because these two variables capture many other variables affected by nature.

To sum it up, only three were kept out of the 20-performance measure extracted from FBREF. The npxG per 90 minutes, the xA per 90 minutes, and the number of pressures on the opposite team per 90 minutes capture players' performance-based market value.

2.3 Empirical Methodology

Multiple OLS[5] regression methods identified vital players' market value components. The regressions were run on a sample constructed of 105 attacking premier league players who had sufficient game time in Europe's top 5 leagues in the last 365 days. The cross-sectional data was updated on April 1st, 2022. Following the regression of Stanojevic & Gyarmati, 2016, the independent variable was the market value exported from Transfermarkt. The first two regressions were based on former academic literature. This paper aims to find an OLS regression model to estimate Premier League attackers' market value. Consequently, an alternative regression is constructed as a third regression.

Multicollinearity must be eliminated to ensure that the regression results are usable. Multicollinearity is a statistical concept where several independent variables in a model are correlated. (Ivestopedia, 2022) If the correlation is high and strong enough between independent variables, it can cause problems when using the regression and interpreting its results. The most common metric is the Variance inflation factors (VIF) to test if multicollinearity exists in a given regression. VIF measures the correlation and the strength of the correlation between the independent variables. Hence, the VIF coefficients are calculated for every regression using STATA[6].

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[5] Least Squares regression (OLS) is a common technique for estimating coefficients of linear regression equations which describe the relationship between one or more independent quantitative variables and a dependent variable (simple or multiple linear regression) (XLSTAT, 2022)

[6] The tests were conducted by using the built-in STATA command named *vif*

Furthermore, the proposed regression models are all tested for heteroscedasticity in STATA[7] using the Breusch-Pagan test; since all economic packages (including STATA) automatically assume homoskedasticity and will calculate the sample variance of the OLS estimator based on the homoskedasticity assumption: $Var(\hat{\beta}) = \sigma^2 (X'X)^{-1}$ (Yamano, 2009)

Thus, in the presence of heteroskedasticity, the statistical inference based on $\sigma^2 (X'X)^{-1}$ would be biased, and t-statistics and F-statistics are inappropriate. To make the results valid if heteroskedasticity exists, the robust standard errors method will be used to adjust the model-based standard errors using the observed variability of the model residuals that are the difference between the observed outcome and the outcome predicted by the statistical model. (Mansournia, 2020)

If we cannot reject the Breusch-Pagan test's null hypothesis, then heteroskedasticity exists, and the regression is re-run using the robust standard errors method in STATA[8], adjusting the model-based standard errors to make the results valid.

Lastly, all regressions are tested in-sample, which means that the regression equations estimate all the in-sample players' market values, and these estimates are then compared to the estimates given by Transfermarkt.

2.3.1 Performance-based regression

The first regression was based on the work of Stanojevic & Gyarmati, 2016, focusing solely on performance measures. The independent variable was the market value, while the dependent variables were the three-performance metrics that proved to be good estimators based on the descriptive statistics.

Equation 1: Performance-based regression's equation

$$Market\ Value = \beta_0 + \beta_1 \times xA + \beta_2 \times npxG + \beta_3 \times Pressures + \epsilon$$

[7] The Breusch-Pagan test is conducted by using the built-in STATA command named *estat het*test

[8] The robust standard-error method is conducted by using the built-in STATA *robust* command

2.1.1 Performance- and personal information-based regression

Analyzing previous academic research and the findings of this paper, the second regression was constructed with the market value as the independent variable again and used three performance-based dependent variables and the three personal information dependent variables introduced below.

Personal information metrics were introduced for the second regression, such as age and contract length. Based on the descriptive statistics, it did not make sense to use any of the personal information metrics as numerical variables in the model. Position and foot variables were dropped as neither had any significant relationship with the market value. Although Kologlu, 2018 found height as a significant component of players' market value, there is no significant relationship between the two variables. Even after factoring the height variable multiple ways, no significant relationship was found.

Following the work of Dobson, S., Howe, S., & Gerrard, B., 2000, a dummy variable out of contract was created. As established in the before-mentioned study, a player's market value should drastically decrease once he enters the last year of his contract. The date of the contract's expiry is almost always June 30th in the Premier League since, in normal circumstances, all domestic and European competitions end by then. Therefore, the out of contract variable takes the value of 1 if a player's contract runs on June 30th, 2022, and 0 otherwise.

Next to other researchers, Li 2020 showed a significant relationship between age and market value. Although no significant relationship was found as a numerical variable, based on Figure 2: Scatter plot graph between market value and age players above 30 years old have significantly lower market value. Therefore, a dummy variable was created called older than 30.

Finally, based on the work of Majewski, 2016, a variable was introduced that captures the teams' value addition to a player's market value. The newly introduced dummy variable works as a control variable for big teams. As top teams, the Big Six[9] was used. These teams are the most valuable

[9] Big Six clubs: Manchester United, Liverpool, Manchester City, Chelsea, Tottenham Hotspurs and Arsenal (Kelly, 2021)

clubs in the Premier League, and 5 of them are even included in the top 10 most valuable clubs in the world. They also have been consistently challenging the top places in the Premier for decades. Hence, all six clubs can rightfully be categorized as top teams, thus constructing the big six variable.

In order to decrease the number of variables, instead of using the Expected Assists and expected non-penalty goals, the variable expected assists + expected non-penalty goals (npxG + xA) is introduced to capture all metrics that are related to a player's potential to score or assist a goal.

Equation 2: Performance- and personal information- based regression's equation

$$\mathrm{M}arket\ Value = \beta_0 + \beta_1 \times (Expected\ Assists + Expected\ Goals) + \beta_2 \times Pressures + \beta_3 \times Out\ of\ Contract + \beta_4 \times Older\ than\ 30 + \beta_5 \times Big\ Six + \epsilon$$

As later shown in the

Empirical Results paragraph, because the out of contract variable's coefficient is not statistically significant, we drop the out of contract variable. However, this shall not disqualify the results of Dobson, Howe, & Gerrard, 2000. The market value defined by Transfermarkt differs from the price a team would need to pay for a player. (Transfermarkt, 2022) While a contract running out would substantially decrease a player's price (essentially making it equal to 0), it does not affect his market value.

Thus, the modified performance- and personal information- based regression equation (

Equation 3) will be used for testing purposes.

Equation 3: Performance- and personal information- based regression modified

$$Market\ Value = \beta_0 + \beta_1 \times npxGxA + \beta_2 \times Pressures + \beta_3 \times Older\ than\ 30 + \beta_4 \times Big\ Six + \epsilon$$

2.1.2 Alternative regression

Next to the five independent variables identified with the performance- and personal information-based regression, the alternative regression proposed by this paper also introduces two new dummy variables, which capture a substantial part of players' market value.

A Premier League club is allowed to register 25 players maximum in its squad for a season. Eight should be homegrown out of the 25 players based on FA rules. A player counts as homegrown if they played at least three years for a club affiliated with the FA before turning 21. So technically, this does not require players to be English; however, most of the players who qualify as homegrown are from England or Wales. (Mukherjee, 2022) Because of the homegrown rule, one would think that English players are generally more valuable for Premier League clubs than non-English players.

Since classifying all 105 players as homegrown or not-homegrown is out of the scope of the research, nationality as a proxy was used instead as multiple professionals would agree that the homegrown rule has indeed inflated the price of English and Welsh players. (Sharma, 2017) Hence, the dummy variable called is English was created, which equals 1 if the player's nationality is either English or Welsh and 0 otherwise.

As already mentioned earlier, none of the academic researchers attempted to include a variable to capture players' commercial potential. However, based on extensive research, professionals agree that players' media and commercial potential significantly impact their market value. A proxy is needed to include the commercial potential of a player. (Athlete Network, 2020) According to Athlete Network, for professional athletes to grow, build, and possibly monetize their brand, Instagram is undoubtedly the absolute best platform. Hence, each player's number of Instagram followers was used.

Based on the Instagram Engagement Report (HubSpot; mention, 2022), only 0.59% of users have more than 1 million followers on Instagram. A dummy variable called Is Star was created to control for "star" players. The dummy takes the value of 1 if a player's Instagram followers are above 1 million and takes the value of 0 if he has less than 1 million followers.

Equation 4: Alternative regression's equation

$$Market\ Value =$$

$$\beta_0 + \beta_1 (Expected\ Assists + Expected\ Goals) + \beta_2 \times Pressures$$
$$+ + \beta_3 \times Older\ than\ 30 + \beta_4 \times Big\ Six + \beta_5 \times is\ English$$
$$+ \beta_6 \times Is\ Star + \epsilon$$

2.1.3 Out of Sample Testing

The out-of-sample (OOS) model validation technique tests the final regression model's efficiency. OOS testing is a methodology created to assess how the results of a linear regression model work on an independent data set.

The out-of-sample test used twenty randomly chosen attackers from the Bundesliga's top 10 clubs (two from each club). Many top-quality attacking players move from the Bundesliga to the Premier League (for example, Timo Werner, Leon Bailey, or Son Heung-min), and since the valuation model is based on Premier League players, Bundesliga attackers are an excellent choice to test the model. Furthermore, the Bundesliga has a similar homegrown player rule as the Premier League: All German teams in the Bundesliga need to have at least eight players who qualify as homegrown players. (Gideon, 2021)

The data was gathered from the same sources as before:

- Market value and personal information: Transfermarkt
- Performance metrics: FBREF
- Commercial potential: Instagram

The final linear regression equation was used to estimate the 20 players' market value, and this market value will be compared to the market values given by Transfermarkt and CIES. The 20 players included in the OOS testing and their extracted data are shown in Appendix 2.

3 Empirical Results

3.1 Performance-based regression

Building on the work of Stanojevic & Gyarmati, 2016, the first regression run tested how well the three-performance metrics can explain the value of attacking players.

3.1.1 Heteroskedasticity test

Table 10: Breusch-Pagan heteroskedasticity test for performance-based regression

Breusch-Pagan / Cook-Weisberg test for heteroskedasticity	
H^0: Constant variance	
Variables: fitted values of MarketValue	
$chi^2(1)$ =	30.12
Prob > chi^2	0.0000

Source: Author

Based on the results of the Breucsh-Pagan test shown in Table 10[10], heteroskedasticity exists. Thus, we ran the regression using the robust method.

3.1.2 Summary of the performance-based regression

Table 11: Results of the performance-based regression, derived from Equation 1

				Number of obs	105	
				F(3, 101)	15.93	
				Prob > F	0.00	
				R-squared	0.4286	
				Root MSE	17.689	
MarketValue	**Coef.**	**Std. Err.**	**t-value**	**P>t**	**[95% Conf. Interval]**	
Expected Assists	177.1	28.3	6.25	0.00	120.85	233.30
Expected Non-Penalty Goals	40.3	16.3	2.47	0.02	7.91	72.59
Pressures	-1.0	0.4	-2.79	0.01	-1.77	-0.30
Constant	6.4	7.6	0.84	0.40	-8.74	21.59

Source: Author

From the performance-based regression in Table 11[11], we can conclude that the performance measures used in the model could capture ~43% of the variation in players' market value. Expected Assists and Expected Non-Penalty Goals positively affect the market value, while Pressures negatively affect it.

[10] Test was conducted using the built-in STATA command named *estat hettest*

[11] Computed with th the built-in STATA command named *robust*

The regression's constant is not significantly different from zero, which means if the three independent variables (Expected Assists, Expected Non-Penalty Goals, and Pressures) are all equal to 0, the player's market value is not significantly different from 0. Since we do not have to consider a case when all our independent variables are equal to 0, we can ignore the constant's significance.

Based on the descriptive statistics and the regression results, the Expected Assists and Expected Non-Penalty Goals proved to have a stronger relationship with players' Market Value than actual Assists and Non-Penalty Goals (detailed explanation in paragraph 2.1). Thus, the Expected Assists and Expected Non-Penalty Goals are used for all regressions in this research.

3.1.3 Multicollinearity test

Table 12: Variance inflation factors (VIF) for the performance-based regression's independent variables

Variable	VIF	1/VIF
Expected Goals	1.09	0.92
Expected Assists	1.08	0.93
Pressures	1.03	0.97
Mean VIF	1.07	

Source: Author

VIF value between 1 and 5 generally indicates a moderate correlation between an independent variable and the other independent variables, but this correlation is not strong enough to give it too much attention. As shown in Table 12, all VIF coefficients are very close to 1. Hence, the regression model is practically free of multicollinearity.

3.1.4 In-sample testing for the performance-based regression

Figure 3: Performance-based regression estimations derived from Equation 1 vs. real market value

Most Valuable Players			
Rank	Name	$\hat{MV}$	MV
1	Mohamed Salah	66.3	100
2	Harry Kane	59.4	100
3	Jack Grealish	52.5	80
4	Heung-min Son	52.1	80
5	Kelechi Iheanacho	50.9	20
6	Edinson Cavani	50.7	4
7	Luis Díaz	50.1	45
8	Timo Werner	49.9	42
9	Raheem Sterling	47.0	85
10	Patson Daka	45.3	22

Least Valuable Players			
Rank	Name	$\hat{MV}$	MV
97	Cenk Tosun	6.4	2.5
98	Jordan Ayew	6.2	6
99	Sergi Canós	5.9	14
100	Francisco Trincão	5.3	22
101	Jóhann Berg Gudmundsson	3.5	3
102	Moussa Djenepo	2.9	11
103	Fábio Silva	2.5	18
104	Przemyslaw Placheta	0.5	2
105	Theo Walcott	0.2	3
106	Aaron Lennon	0.0	1

Source: Author

Based on performance metrics, out of the 105 players in the sample, Mohammed Salah from Liverpool is the most valuable attacking player in the Premier League, with €66 million, while Aaron Lennon from Burnley with €0 million. Both the most valuable and the least valuable player matches Transfermarkt's most valuable and least valuable player. As expected, with an R^2 of 0.43, the market value estimations are not so accurate. However, we can see that with some exceptions, the relative market value of players is quite accurate.

To fine-tune the regression model, three new variables were introduced in the second regression, with which the intention was to control for specific player characteristics which are not performance-related, such as age or expiring contracts.

3.2 Results of the performance- and personal info-based regression

The second regression was meant to replicate the first regression while also controlling for the effect of ending contracts, players' age, and the anomaly caused by the prestige of the players club.

3.2.1 Heteroskedasticity test

Table 13: Breusch-Pagan heteroskedasticity test for the performance and personal information-based regression

Breusch-Pagan / Cook-Weisberg test for heteroskedasticity	
H^0: Constant variance	
Variables: fitted values of MarketValue	
$chi^2(1)$ =	40.79
Prob > chi^2	0.0000

Source: Author

Based on the results of the Breucsh-Pagan test shown in Table 13, heteroskedasticity exists. Thus, we ran the regression using the robust method.

3.2.2 Summary of the performance- and personal info-based regression

Table 14: Results of performance- and personal info-based regression, derived from Equation 2

				Number of obs	105	
				F(5, 99)	16.3	
				Prob > F	0.00	
				R-squared	0.5893	
				Root MSE	14.892	
MarketValue	**Coef.**	**Std. Err.**	**t-value**	**P>t**	**[95% Conf. Interval]**	
Expected Assists + Goals	28.5	14.4	1.98	0.05	-0.12	57.05
Pressures	-1.0	0.3	-3.00	0.00	-1.72	-0.35
Out of Contract	-6.0	3.4	-1.77	0.08	-12.67	0.73
Older than 30	-14.0	2.6	-5.36	0.00	-19.13	-8.79
Big Six	25.1	5.2	4.85	0.00	14.81	35.35
Constant	28.0	8.2	3.41	0.00	11.70	44.30

Source: Author

The three new variables highly improve the regression model[12]. Four are statistically significant out of the five independent variables used for the regression. However, the newly introduced Out of Contract variable's coefficient is insignificant, with a 5% confidence level (see paragraph 2.2.1).

[12] Computed with th the built-in STATA command named *robust*

Regarding the control variables introduced in the performance- and personal info-based regression, all three of them have a significant effect on the Market Value:

- Out of contract: As per Dobson, S., Howe, S., & Gerrard, B. (2000), players with a contract ending within the next year suffer an average Market Value decrease of €6 million. Still, the coefficient is statistically not significant. Hence, the variable is dropped.
- Older than 30: As one would suspect after Li, 2020's work, 30 years old or older players' Market Value is on average €14 million less than players from a similar club with equal performance metrics and contract conditions.
- Big Six: Finally, the regression confirms the findings of Majewski, 2016 as well. Players from top teams have a higher Market Value of €25 million on average, compared to players with similar metrics, but not from a top club. However, the Big Six variable's effect is expected to decrease after introducing the variable that captures commercial potential since these teams tend to have more star players in their squad.

3.2.3 Multicollinearity test

Table 15: Variance inflation factors for the performance- and personal information-based regression's independent variables

Variable	**VIF**	**1/VIF**
Expected Assists + Goals	1.59	0.63
Big Six	1.58	0.63
Out of Contract	1.17	0.85
Older than 30	1.16	0.86
Pressures	1.05	0.95
Mean VIF	1.31	

Source: Author

VIF value between 1 and 5 generally indicates a moderate correlation between an independent variable and the other independent variables, but this correlation is not strong enough to give it too much attention. As shown in Table 12, all VIF coefficients are very close to 1. Hence, the regression model is practically free of multicollinearity.

3.2.4 In-sample testing for performance- and personal info-based regression

As mentioned above, the Out of Contract variable's coefficient has no significance. Thus the equation used to compare the estimates of the performance- and personal info-based regression and Transfermarkt's estimates is

Equation 3.

Table 16: Comparing Performance- and personal info-based regression estimations derived from Equation 3 to Transfermarkt estimations

Most Valuable Players			
Rank	**Name**	$\widehat{MV}$	**MV**
1	Harry Kane	63.6	100
2	Mohamed Salah	61.7	100
3	Timo Werner	59.4	42
4	Romelu Lukaku	58.7	85
5	Raheem Sterling	58.6	85
6	Heung-min Son	58.5	80
7	Sadio Mané	58.4	80
8	Cristiano Ronaldo	57.6	35
9	Edinson Cavani	56.0	4
10	Gabriel Jesus	54.8	50

Least Valuable Players			
Rank	**Name**	$\widehat{MV}$	**MV**
97	Anthony Gordon	12.3	16
98	Przemyslaw Placheta	12.0	2
99	Ashley Barnes	11.2	2
100	Teemu Pukki	11.2	5
101	Alex Iwobi	10.7	18
102	Andros Townsend	10.0	7
103	Nathan Tella	7.9	2
104	Theo Walcott	6.2	3
105	Jordan Ayew	5.1	6
106	Aaron Lennon	2.6	1

Source: Author

Based on the performance- and personal info-based regression, out of the 105 players in the sample, Harry Kane from Tottenham is the most valuable attacking player (same as in the Transfermarkt database) in the Premier League, while Aaron Lennon is again from Burnley, the least valuable player (same as in TM database). The relative market values are pretty accurate again. However, looking at the table indicates that the standard deviation is more considerable for the values estimated by Transfermarkt.

3.3 Results of the alternative regression

3.3.1 Heteroskedasticity test

Table 17: Breusch-Pagan heteroskedasticity test

Breusch-Pagan / Cook-Weisberg test for heteroskedasticity	
H^0: Constant variance	
Variables: fitted values of MarketValue	
$chi^2(1)$ =	39.44
Prob > chi^2	0.0000

Source: Author

The Breusch-Pagan test was run in STATA[13] to test the regression for heteroskedasticity. As shown in Table 17, we cannot reject the null hypothesis, which means heteroskedasticity exists.

3.3.2 Summary of the alternative regression model

The alternative regression uses the robust standard errors method in STATA[14] to validate the results, which essentially adjusts the model-based standard errors.

[13] Test was conducted using the built-in STATA command named *estat hettest*

[14] Computed with th the built-in STATA command named *robust*

Table 18: Alternative regression model's results with robust, derived from Equation 4

				Number of obs	105	
				F(6 , 98)	17.59	
				Prob > F	0.00	
				R-squared	0.6529	
				Root MSE	0	
MarketValue	**Coef.**	**Std. Err.**	**t-value**	**P>t**	**[95% Conf. Interval]**	
Expected Assists + Goals	32.1	13.7	2.34	0.02	4.92	59.34
Pressures	-1.1	0.3	-3.33	0.00	-1.70	-0.43
Older than 30	-16.7	3.2	-5.20	0.00	-23.01	-10.30
Big Six	12.7	6.3	2.03	0.05	0.27	25.09
Is English	7.0	3.0	2.35	0.02	1.09	12.91
Is Star	16.1	4.8	3.37	0.00	6.61	25.60
Constant	22.4	7.6	2.94	0.00	7.27	37.46

Source: Author

After implementing the robust standard error method, we have our final linear regression to estimate the attackers' market value. The alternative model based on Equation 4: Alternative regression's equation captures ~65% of the variation in player's market value. All the coefficients are statistically significant based on the T-tests, while the regression is also significant by looking at the F-statistic.

3.3.3 Multicollinearity test

Table 19: Variance inflation factor for all independent variables

Variable	**VIF**	**1/VIF**
Expected Assists + Goals	2.81	0.36
Pressures	2.22	0.45
Older than 30	1.61	0.62
Big Six	1.04	0.96
Is English	1.02	0.98
Is Star	1.02	0.98
Mean VIF	1.62	

Source: Author

The VIF was calculated in STATA to test the regression for multicollinearity. A VIF value between 1 and 5 generally indicates a moderate correlation between an independent variable and the other independent variables, but this correlation is not strong enough to give it too much attention. Thus, the regression model is practically free of multicollinearity.

3.3.4 In-sample testing for the alternative regression

Table 20: Alternative regression model's estimations derived from Equation 4 vs. market value estimates from Transfermarkt and CIES

Most Valuable Players				
Rank	**Name**	$\widehat{MV}$	**TM MV**	**CIES MV**
1	Harry Kane	70.6	100	50-70
2	Raheem Sterling	65.1	85	50-70
3	Cristiano Ronaldo	63.5	35	15-20
4	Mohamed Salah	61.8	100	50-70
5	Jack Grealish	59.7	80	50-70
6	Timo Werner	59.3	42	50-70
7	Romelu Lukaku	58.1	85	50-70
8	Sadio Mané	58.1	80	40-50
9	Heung-min Son	58.0	80	40-50
10	Callum Hudson-Odoi	56.7	30	40-50
11	Jadon Sancho	56.3	70	90-120
12	Gabriel Jesus	54.0	50	50-70
13	Riyad Mahrez	53.8	40	30-40
14	Luis Díaz	53.1	45	50-70
15	Marcus Rashford	52.5	80	70-90
16	Diogo Jota	52.5	60	70-90
17	Nicolas Pépé	50.3	30	20-30
18	Gabriel Martinelli	50.1	38	30-40
19	Roberto Firmino	47.5	38	20-30
20	Jamie Vardy	47.4	5	5-7

Source: Author; Adapted from: Transfermarkt.com (2022), CIES (2022)

Based on the alternative regression model, out of the 105 players in the sample, Harry Kane is the most valuable attacking player in the Premier League, which is in line with Transfermarkt's estimates but differs from CIES's values. Interestingly, the estimated market values for the alternative model's top 20 players are more in-line with CIES's database than with Transfermarkt's.

The five most overvalued players[15] based on the alternative regression estimates and market values provided by professional data providers are:

- Edinson Cavani (Manchester United),
- Jamie Vardy (Leicester City),
- Theo Walcott (Everton),
- Christian Benteke (Crystal Palace),
- Jóhann Berg Gudmundsson (Burnley).

Interestingly enough, all of the players mentioned above are 31 years old or older, and all four of them were once crucial players for their respective clubs.

The five most undervalued players[16] based on the alternative regression estimates and market values provided by professional data providers occurred the following players:

- Francisco Trincão (Wolverhampton),
- Hee-chan Hwang (Wolverhampton),
- Jordan Ayew (Crystal Palace),
- Teemu Pukki (Norwich City),
- Aaron Lennon (Burnley).

Besides the two Wolverhampton players, these players are all aged 30 years old or more. Thus, based on the relative differences between the estimated market value and the market values given by Transfermarkt, it seems like the model is less reliable for players above 30, even though a control variable was included for this specific group of players. As shown in Table 20, Cristiano Ronaldo (37 years old) is also overvalued by the alternative regression model.

---

[15] Highest relative difference: Estimated market value / Transfermarkt's market value

[16] Lowest relative difference: Estimated market value / Transfermarkt's market value

3.4 Out-of-sample testing

3.4.1 Modifications made for the OOS testing

The performance-based equation does not need any modification, as there is no Premier League's specific variable included in its equation. Hence, Equation 1 is kept for the out-of-sample testing.

The Big Six variable does not make sense in the Bundesliga for the performance- and personal information-based regression. However, in recent years, three clubs have consistently outperformed the rest: Bayern München, Dortmund, and Leipzig. As regular Champions League clubs, they have a larger fanbase domestically and internationally, making them the big three clubs in Germany. Thus, the dummy Big Six is dropped for the OOS testing, and the dummy Big Three is introduced. Big Three takes the value of 1 if a player is from one of the clubs mentioned above and 0 otherwise. Hence, instead of using Equation 2, Equation 5 is used for the OOS testing.

Equation 5: Performance- and personal information- based regression equation for the Bundesliga

$$\begin{aligned} &Market\ Value = \\ &\beta_0 + \beta_1 \times (Expected\ Assists + Expected\ Goals) + \beta_2 \times Pressures \\ &\qquad + \beta_3 \times Out\ of\ Contract + \beta_4 \times Older\ than\ 30 + \beta_5 \times Big\ Three + \epsilon \end{aligned}$$

Furthermore, the alternative regression model's equation has to be changed because of the Is English variable. As earlier mentioned, the Bundesliga has a very similar homegrown rule as the Premier League, favoring German players. Thus, for the OOS testing, Is English is changed to Is German. Is German takes the value of 1 if a player's first nationality is German and takes 0 otherwise. For the testing,

Equation 6 is used instead of Equation 4.

Equation 6: Alternative model's equation altered to Bundesliga players

$$Market\ Value = \beta_0 + \beta_1 \times (Expected\ Assists + Expected\ Goals) + \beta_2 \times NP\ Goals + \beta_3 \times Pressures + \beta_5 \times Older\ than\ 30 + \beta_6 \times Big\ Three + \beta_7 \times is\ German + \beta_6 \times Is\ Star$$

All other variables are kept the same as they represent the same concept in the Premier League and the Bundesliga.

3.4.2 Out-of-sample testing results

Table 21: Alternative regression's OOS estimations vs. real market value

Most Valuable Players						
		Performance-based model	Performance- and personal information-based	Alternative model	Transfermarkt	CIES
Name	**Club**	$\hat{MV}$	$\hat{MV}$	$\hat{MV}$	$\hat{MV}$	$\hat{MV}$
Leroy Sané	Bayern München	81.2	57.7	63.9	70	70-90
Thomas Müller	Bayern München	60.6	48.0	43.0	25	20-30
Thorgan Hazard	Dortmund	29.9	42.3	23.5	20	15-20
Donyel Malen	Dortmund	51.0	54.0	36.4	27	50-70
Paulinho	Bayer Leverkusen	51.3	25.4	19.9	11	4-7
Patrick Shick	Bayer Leverkusen	113.0	37.4	33.3	40	30-40
Roland Sallai	Freiburg	60.2	27.6	22.4	10	10-15
Lucas Höler	Freiburg	43.0	21.3	22.3	6.5	7-10
Emil Forsberg	RB Leipzig	39.1	42.1	13.2	16	20-30
Yussuf Poulsen	RB Leipzig	51.1	43.9	25.4	21	15-20
Jan Thielman	Köln	8.4	13.0	13.1	5.5	7-10
Mark Uth	Köln	41.4	12.8	3.0	3	2-4
Taiwo Awoniyi	Union Berlin	82.7	29.1	24.1	15	15-20
Andreas Voglsammer	Union Berlin	46.6	14.9	5.2	1.2	NA
Ihlas Bebou	Hoffenheim	58.2	25.0	26.4	16	10-15
Jacob Bruun Larsen	Hoffenheim	55.5	27.5	22.2	6.5	4-7
Jonathan Burkardt	Mainz	76.0	28.1	29.9	17	20-30
Karim Onisiwo	Mainz	54.2	17.6	1.3	3.5	4-7
Alassane Plea	Borussia Mönchengladbach	50.8	29.2	24.1	15	7-10
Breel Embolo	Borussia Mönchengladbach	53.8	29.2	24.2	14	15-20

Source: Author; Adapted from: Transfermarkt.com (2022), CIES (2022)

Equation 6 was used to estimate the selected Bundesliga players' market value. The estimations were compared to Transfermarkt's (TM) and CIES's estimates.

As shown in Table 21, regression performs better than expected for Bundesliga players. The four players for whom the estimated Market Value was very different from both Transfermarkt and CIES estimates are:

- Thomas Müller (Bayern München),
- Paulinho (Bayer Leverkusen),
- Lucas Höler (Freiburg),
- Jacob Bruun Larsen (Hoffenheim).

Again, two players out of the four are above 30 years old (Thomas Müller and Lucas Höler), implying that the model could be further improved by creating a more sophisticated age variable rather than only a dummy variable.

3.4.3 Mean-square analysis

The mean squared error (MSE) estimator assesses whether the alternative regression model has better predicting power than the regressions based on academic literature.

The MSE represents the average squared difference between the estimated values and the actual value for each regression in the following way:

Equation 7: Mean square error's equation

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y_i})^2$$

Source: Wikipedia (2022), https://en.wikipedia.org/wiki/Mean_squared_error#Interpretation

Equation 7 is used to calculate the MSE coefficients for all three regression models used for the OOS testing, namely:

- Performance-based regression model,
- Performance- and personal information-based regression model,
- Alternative regression model.

As all the models were based on Transfermarkt's market values, the expected value (Y_i) is always the respective player's Transfermarkt market value estimation.

Table 22: Regression models' MSE coefficients

Regression model	MSE
Performance-based	1 775.9
Performance- and personal information-based	283.2
Alternative	94.3

Source: Author

As shown in Table 22, the alternative model has the lowest MSE, which is the best model out of the three to estimate attacking players' market value.

4 Model limitations and thoughts on future improvements

As with all models, the alternative regression model introduced in this paper is just a simplified version of the actual situation. The world of football is a very complex system in which countless variables influence the value of players. The number of assumptions made in the model causes a difference between the model and reality. However, creating models is an easy and cheap way to get an idea about some real-world interrelations. Keeping this in mind, this paper has some specific limitations. Addressing these limitations with the proposed improvements could increase the model's estimation power.

4.1 Lack of historical data

As mentioned before, data is precious in the football industry. Therefore, there are no freely accessible databases that let researchers screen data efficiently. Professional companies such as Hudl/Wyscout or Statsbomb charge thousands of euros for database access. Unfortunately, based on a personal experience from several emails and meetings with these firms, they do not offer free access to students. On the other hand, accessible databases, even if they have a decent amount of data, like FBREF, do not let users easily export a mass amount of data. These databases are primarily created for enthusiastic football individuals instead of researchers. Therefore, the only way to efficiently gather data from these websites is to write web scraping codes.

Getting access to a professional data provider would make it possible to build a panel database instead of a cross-sectional data sample. Using a panel[17] database instead of a cross-sectional would highly increase the sample size, allowing the paper to check the yearly effects or introduce lagged data.

4.2 Transfermarkt's market value instead of actual transfer values

Transfermarkt does not use an algorithm but instead relies on the wisdom of the community. Although the wisdom of the community could be seen as a market consensus, since Transfermarkt has one of the largest active football communities in the world, there is no scientific approach behind its method. It is also important to note that the market value calculated by Transfermarkt is not equal to the price. This makes the contract length variable introduced redundant. An expiring contract makes a player's price equal to zero; however, it does not affect the player's market value. (Transfermarkt, 2022)

To improve the model, the independent variable could be changed from TM's market value to actual historical transfer prices. However, building such a database would require a professional database, as a player's metrics must be valid at the transfer's date. A regression model built on actual transfer prices would allow us to bring a player's market value closer to its price.

4.3 Expected vs. actual performance measures values

Both goals and assists have been included in the regression model using their expected measures. For this research, twenty different performance measures were exported from FBREF. Nevertheless, these measures have high collinearity, so the number of variables had to be decreased. As seen in the Descriptive statistics (paragraph 2.2), expected goals and expected assists are highly correlated with almost all other attacking performance measures. This most likely comes from the expected measures calculated using other performance statistics. However, the exact methodology of calculating the expected goals and expected assists is not disclosed by FBREF.

[17] Panel data (or time series cross section) means that we have data from many units, over many points in time

One way to improve the model and have less exposure to FBREF's methodology would be to use the principal component analysis method, which would let us transform the large set of performance variables into a smaller one that still contains most of the information. This way, the summarized variables would include all the information from the different performance measures instead of relying on the expected goals and expected assists calculated by FBREF, which essentially does the same without letting us know how these variables are composed precisely.

5 Summary

The paper aimed to contribute to the field of football player valuation with a regression model that can be used to estimate attacking football players' market value. Using cross-sectional data including 105 attacking players from the English Premier League, 27 metrics exported from three databases were evaluated as potential variables.

Two different linear regression models based on academic literature were evaluated. While improving the regression models based on the prior academic research, this paper successfully introduced an alternative regression that can be used to estimate attacking football players' value more precisely.

The alternative regression model introduced by this research had an R^2 of 0.65 in-sample, which was higher than the performance-based and the performance- and personal information-based regression model's $R^{2,}$ both based on prior academic literature. The alternative model also outperformed the two other models in the out-of-sample testing, with the lowest mean-square estimator coefficient (MSE = 94.3).

According to the findings presented in this research, the following variables have a significant effect on a player's market value:

- Expected goals + expected assists per 90 min (in the last 365 days),
- Pressures per 90 min (in the last 365 days),
- Player's age (whether the player is above 30 years old or not),
- Player's nationality (whether a player's nationality is the same as the league's),
- Player's club's status (whether a player plays in a prestigious club or not),

- Player’s commercial potential (whether the player has more than 1m Instagram followers or not)

The regression introduced by the paper identified two new variables, (1) commercial potential and (2) nationality, which were not included in other academic papers, but had a statistically significant impact on the market value. A player’s commercial potential was proxied by his Instagram follower numbers: players with more than 1 million followers have a statistically significant higher market value than players with fewer followers (all else equal). Moreover, the anomaly caused by the homegrown rules, players from the league’s nationality have a significantly higher market value than players with similar characteristics but from another nation.

Although the paper softened some data-gathering limitations by using a web scraping method, this study still has limitations. However, using a professional data provider for a similar study could provide an opportunity to overcome the majority of these limitations, with the ability to build up a panel data sample and perform a more in-depth analysis on the topic.

Bibliography

Accounting Tools. (2022). *accountingtools.com*. Retrieved from What is Human Capital?: https://www.accountingtools.com/articles/what-is-human-capital.html#:~:text=The%20value%20of%20human%20capital,but%20rather%20by%20its%20employees.

Ace Advisory. (2022). *Ace Advisory*. Retrieved from methodology: https://footballbenchmark.com/methodology/player_valuation

AI Abacus. (2022). *About Us*. Retrieved from Ai Abacus: https://ai-abacus.co.uk/about-us/

Athlete Network. (2020). *Athlete Network*. Retrieved from Here's The One Social Media Platform That Every Athlete Should Use: https://an.athletenetwork.com/blog/social-media-platform-every-athlete-needs#:~:text=If%20you%20are%20a%20professional,platform%20you%20should%20be%20on.

Cam, I., & Özer, G. (2016). The Role of Human Capital in Firm Valuation: An Application on. *12th International Strategic Management Conference*, (pp. 168 – 177).

CIES. (2020). *CIES*. Retrieved from Scientific evaluation of the transfer value of football players: https://www.football-observatory.com/IMG/sites/mr/mr53/en/

CIES. (2022). *CIES*. Retrieved from https://football-observatory.com/

Dobson, S., Howe, S., & Gerrard, B. (2000). The determination of transfer fees in English nonleague football. *Applied Economics*, 1145-1152.

Edward Nsolo, P. L. (2018). *Player Valuation in European Football* .

FBREF. (2022). *FBREF*. Retrieved from xG Explained: https://fbref.com/en/expected-goals-model-explained/

FBREF. (2022). *FBREF*. Retrieved from About: https://fbref.com/en/

Gamerschlag, R. (2013). Value relevance of human capital information. *Journal of Intellectual Capital*.

Gideon. (2021). *Football Handbook*. Retrieved from The Ultimate Guide To Homegrown Players: https://footballhandbook.com/homegrown-player/#:~:text=at%208%20players.-,What%20is%20the%20homegrown%20player%20rule%20in%20Germany%3F,at%20least%208%20qualifying%20players.

Haigh, R. (2017). *brandirectory*. Retrieved from GIFT report 2016: https://brandirectory.com/reports/global-intangible-financial-tracker-gift-2016

Hopkins, F. (2021). *frankphopkins*. Retrieved from Scraping FBRef to perform comparative football player analysis: https://frankphopkins.medium.com/scraping-fbref-to-perform-comparative-football-player-analysis-cb8a1a2675a6

HubSpot; mention. (2022). *Instagram Engagement Report, 202 edition.*

Investopedia. (2022). *Investopedia*. Retrieved from Human Capital: https://www.investopedia.com/terms/h/humancapital.asp

Ivestopedia. (2022). *Investopedia*. Retrieved from Multicollinearity: https://www.investopedia.com/terms/m/multicollinearity.asp#:~:text=Multicollinearity%20is%20a%20statistical%20concept,in%20less%20reliable%20statistical%20inferences.

Jose Luis Felipe, A. F.-L.-S.-U. (2020). *Team Variables and Player Positions that Most Influence the Market Value of Professional Male Footballers in Europe.*

Kelly, R. (2021). *Goal.com*. Retrieved from Who are the Premier League' big six'?: https://www.goal.com/en-us/news/who-are-premier-league-big-six-top-english-clubs-nickname/130iokmi8t8dt1k3kudou73s1k

Kologlu, Y., Birinci, H., Kanalmaz, S., & Özyilmaz, B. (2018). *A Multiple Linear Regression Approach For Estimating the Market Value of Football Players in Forward Position.*

Li, S. (2020). Multiple regression model for predicting soccer player value in English Primer League. *The Frontiers of Society, Science and Technology*, 132-143.

Likert, R. (1971). Human Resource Accounting: A Human Organizational Measurement Approach. *Financial Analyst Journal*, 75-84.

Majewski, S. (2016). Identification of Factors Determining Market Value of the Most Valuable Football Players. *Journal of Management and Business Administration. Central Europe*, 91-104.

Mansournia, M. A. (2020). Reflection on modern methods: demystifying robust standard errors for epidemiologists. *International Journal of Epidemiology*, 346–351.

Merriman, K. K. (2017). *Valuation of Human Capital.*

Metelski, A. (2021). Factors affecting the value of football players in the transfer market. *Journal of Physical Education and Sport*, 1150-1155.

Miao He, R. C. (2015). *Football Player's Performance and Market Value.*

Milost, F. (2013). *Models of Human Capital Valuation.*

Mukherjee, S. (2022). *Goal.com*. Retrieved from How many foreign players can a Premier League club sign?: https://www.goal.com/en-in/news/how-many-foreign-players-can-a-premier-league-club-sign/bltb28934959d4e82d3

Nicolas Leifheit, F. F. (2021). Financial player valuation from the perspective of. *Managing Sport and Leisure*.

Our World in Data. (2022). *Our World in Data*. Retrieved from Human Capital Index vs. GDP per capita: https://ourworldindata.org/grapher/human-capital-index-vs-gdp

Pelinescu, E. (2014). *The impact of human capital on economic growth.*

Perechuda, I., & Cater, T. (2021). *Influence of stakeholders'perception on value creation and measurement: the case of football clubs.*

Rob A. Wilson, G. B. (2004). *The impact of human capital.*

RSM. (2017). *RSM*. Retrieved from Football Stars as Intangible Assets: https://www.rsm.global/indonesia/en/insights/articles/football-stars-intangible-assets

Sharma, H. (2017). *90min*. Retrieved from Why are Egnlish players so expensive?: https://www.90min.com/in/posts/5296609-why-are-english-footballers-so-expensive-we-give-you-four-reasons-which-explain-the-trend

Stanojevic, R., & Gyarmati, L. (2016). Towards data-driven football player assessment. *16th International Conference on Data Mining Workshops.*

Transfermarkt. (2020). *Transfermarkt*. Retrieved from https://www.transfermarkt.com/

Transfermarkt. (2022). *Transfermarkt*. Retrieved from Transfermarkt Market Value explained - How is it determined?: https://www.transfermarkt.co.in/transfermarkt-market-value-explained-how-is-it-determined-/view/news/385100

Understat. (2022). *Understat*. Retrieved from https://understat.com/

Weisbrod, B. A. (1961). The Valuation of Human Capital. *Journal of Political Economy*, 425-436.

Wicker, P. (2013). No Pain, No Gain: Effort and Productivity in Professional Soccer. *International Journal of Sport Finance*.

Wikipedia. (2022). *Wikipedia*. Retrieved from Ballon d'Or: https://en.wikipedia.org/wiki/Ballon_d%27Or

Wikipedia. (2022). *Wikipedia*. Retrieved from The Football Association: https://en.wikipedia.org/wiki/The_Football_Association

Wikipedia. (2022). *Wikipedia*. Retrieved from Ballon d'Or: https://en.wikipedia.org/wiki/Ballon_d%27Or

Wikipedia. (2022). *Wikipedia*. Retrieved from Python (programming language): https://en.wikipedia.org/wiki/Python_(programming_language)

Wikipedia. (2022). *Wikipedia*. Retrieved from multicollinearity: https://en.wikipedia.org/wiki/Multicollinearity

World Economy Forum. (2020). *WEFORUM*. Retrieved from We need to get better at measuring the value of human capital - here's how to do it: https://www.weforum.org/agenda/2020/10/why-we-need-a-new-human-capital-accounting-framework-and-how-to-get-there/

XLSTAT. (2022). *XLSTAT*. Retrieved from ORDINARY LEAST SQUARES REGRESSION (OLS): https://www.xlstat.com/en/solutions/features/ordinary-least-squares-regression-ols

Yamano, T. (2009). *Heteroskedasticity and Robust Estimators* .

Yunhu Kim, K.-H. N. (2019). *Data-driven exploratory approach on player valuation in football transfer market.* Chung-Ang University.

Appendix

Appendix 1: Web scraping code in Python

```
[ ] import requests
    import xlrd
    import pandas as pd
    from bs4 import BeautifulSoup
    import klib as kb
    import seaborn as sb
    import matplotlib.pyplot as plt
    import wes
    import matplotlib as mpl
    import warnings
    import numpy as np
    from math import pi
    from urllib.parse import unquote
```

```
[ ] from google.colab import files
    uploaded = files.upload()
```

```
[ ] def get_player_data(x):
        warnings.filterwarnings("ignore")
        url = x
        page =requests.get(url)
        soup = BeautifulSoup(page.content, 'html.parser')
        name = [element.text for element in soup.find_all("span")]
        name = name[7]
        metric_names = []
        metric_values = []
        remove_content = ["'", "[", "]", ","]
        for row in soup.findAll('table')[0].tbody.findAll('tr'):
            first_column = row.findAll('th')[0].contents
            metric_names.append(first_column)
        for row in soup.findAll('table')[0].tbody.findAll('tr'):
            first_column = row.findAll('td')[0].contents
            metric_values.append(first_column)
        metric_values = repr(metric_values)
        for content in remove_content:
            metric_values = metric_values.replace(content, '')
        metric_values = metric_values.split(" ")

        if len(metric_values) < 21:
          metric_values=['NA', 'NA', 'NA',
                         'NA', 'NA', 'NA',
                         'NA', 'NA', 'NA',
                         'NA', 'NA', 'NA',
                         'NA', 'NA', 'NA',
                         'NA', 'NA', 'NA',
                         'NA', 'NA', ]

        else:
          metric_values.pop(7)
          metric_values.pop(14)

        return(metric_values)
```

```
[ ] wb = xlrd.open_workbook("Database.xlsm")
    sheet = wb.sheet_by_index(1)
```

```
[ ] df_player = pd.DataFrame(metric_values_all)
    df_player.columns = ['Non-Penalty Goals', 'npxG',
                         'Shots Total', 'Assists', 'xA',
                         'npxG+xA', 'Shot-Creating Actions',
                         'Passes Attempted', 'Pass Completion %',
                         'Progressive Passes', 'Progressive Carries',
                         'Dribbles Completed', 'Touches (Att Pen)',
                         'Progressive Passes Rec', 'Pressures',
                         'Tackles', 'Interceptions', 'Blocks',
                         'Clearances', 'Aerials won' ]
    df_player.index = player_names_list
```

```
[ ] metric_values_all=[]
    j = 0
    while j < len(url_list):
      metric_values_all.append(get_player_data(url_list[j]))
      j = j +1
```



```
[ ] file_name = 'fbref_dataset.xlsx'

    df_player.to_excel(file_name)
```

Appendix 2: Data extracted for Bundesliga players

Name	Club	npxG	xA	npxG + xA	Pressures	Older than 30	Big Three	Is German	Is Star
Leroy Sané	Bayern München	0.45	0.28	0.73	16.16	0	1	1	1
Thomas Müller	Bayern München	0.32	0.39	0.67	18.17	1	1	1	1
Thorgan Hazard	Dortmund	0.17	0.35	0.35	20.73	0	1	0	0
Donyel Malen	Dortmund	0.28	0.18	0.46	12.26	0	1	0	0
Paulinho	Bayer Leverkusen	0.31	0.14	0.46	15.69	0	0	0	0
Patrick Shick	Bayer Leverkusen	0.65	0.05	0.7	10.52	0	0	0	0
Roland Sallai	Freiburg	0.34	0.21	0.51	14.92	0	0	0	0
Lucas Höler	Freiburg	0.28	0.14	0.42	18.68	0	0	1	0
Emil Forsberg	RB Leipzig	0.24	0.18	0.42	17.02	1	1	0	0
Yussuf Poulsen	RB Leipzig	0.35	0.16	0.51	23.73	0	1	0	0
Jan Thielman	Köln	0.1	0.18	0.28	22.98	0	0	1	0
Mark Uth	Köln	0.28	0.24	0.53	24.27	1	0	1	0
Taiwo Awoniyi	Union Berlin	0.5	0.07	0.57	15.11	0	0	0	0
Andreas Voglsammer	Union Berlin	0.31	0.13	0.45	19.94	1	0	1	0
Ihlas Bebou	Hoffenheim	0.36	0.11	0.47	16.43	0	0	1	0
Jacob Bruun Larsen	Hoffenheim	0.32	0.18	0.5	14.8	0	0	0	0
Jonathan Burkardt	Mainz	0.46	0.1	0.56	15.89	0	0	1	0
Karim Onisiwo	Mainz	0.33	0.24	0.56	20.35	1	0	0	0
Alassane Plea	Borussia Mönchengladbach	0.27	0.31	0.6	15.93	0	0	0	0
Breel Embolo	Borussia Mönchengladbach	0.32	0.2	0.65	17.35	0	0	0	0

Source: FBREF; Date extracted: 2022/05/01

Appendix 3: Do-file used in STATA

Thesis_do file_final*

```

*DESCRIPTIVE STATISTICS

	*descriptive statistics of Market Value, Height and Contract Length
	sum MarketValue Age Heightcm ContractLength

	*correlation between Market Value and personal information metrics
	corr MarketValue Age Heightcm ContractLength

	*descriptive statistics for number of followers
	sum Followers000, detail

	*correlation between Market Value and performance measures
	corr MarketValue Non_Penalty_Goals npxG ShotsTotal Assists xA npxGxA ///
	ShotCreatingActions PassesAttempted PassCompletion ProgressivePasses ///
	ProgressiveCarries DribblesCompleted TouchesAttPen ProgressivePassesRec ///
	Pressures Tackles Interceptions Blocks Clearances Aerialswon ///

*PERFORMANCE-BASED REGRESSION

	*results of performance-based regression
	reg MarketValue xA npxG Pressures

	*Breusch-Pagan heteroskedasticity
	estat hettest

	*results of performance-based regression, with robust
	reg MarketValue xA npxG Pressures, robust

	*variance inflation factor for all independent variables
	vif

*PERFORMANCE- AND PERSONAL INFO-BASED REGRESSION

	*results of performance- and personal info-based regression
	reg MarketValue npxGxA Pressures Out_of_Contract Above_30_years_old big6

	*Breusch-Pagan heteroskedasticity
	estat hettest

	*results of performance- and personal info-based regression
	reg MarketValue npxGxA Pressures Out_of_Contract Above_30_years_old ///
	big6, robust

	*variance inflation factor for all independent variables
	vif


*ALTERNATIVE REGRESSION

	*results of the alternative regression
	reg MarketValue npxGxA Pressures Above_30_years_old big6 isEnglish isStar

	*Breusch-Pagan heteroskedasticity
	estat hettest

	*results of the alternative regression with robust
	reg MarketValue npxGxA Pressures Above_30_years_old big6 isEnglish ///
	isStar, robust

	*variance inflation factor for all independent variables
	vif

```